

2021-22 Virtual Challenge Meet #2 – Science Test

ANSWER KEY

B01. A	C01. B	P01. D
B02. B	C02. E	P02. B
B03. D	C03. D	P03. E
B04. D	C04. A	P04. C
B05. C	C05. A	P05. C
B06. C	C06. C	P06. A
B07. A	C07. C	P07. C
B08. C	C08. B	P08. D
B09. A	C09. B	P09. E
B10. D	C10. D	P10. B
B11. A	C11. E	P11. E
B12. B	C12. C	P12. A
B13. E	C13. E	P13. C
B14. D	C14. E	P14. A
B15. A	C15. C	P15. D
B16. A	C16. E	P16. E
B17. B	C17. A	P17. E
B18. B	C18. A	P18. B
B19. B	C19. D	P19. C
B20. C	C20. A	P20. D

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SELECTED SOLUTIONS

C01. HNO_3

C03. $4 \text{NH}_3 + 3 \text{O}_2 \rightarrow 2 \text{N}_2 + 6 \text{H}_2\text{O}$ ammonia is limiting, $4.5 \text{ mol}(2/4) = 2.25 \text{ L}$ nitrogen produced

C04. $4.560 + 0.0026 = 4.563$

C07. $q = (1.52 \text{ kJ/K})(1.73\text{K}) = 2.63 \text{ kJ}$ $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4$ $(0.100\text{L})(1.00 \text{ M}) = 0.100 \text{ mol}$

Enthalpy chg = -2.63 kJ 0.100 mol BaSO_4 produced $-2.63/0.100 = -26.3 \text{ kJ/mol}$

C08. $4 \text{Bi(s)} + 3 \text{O}_2(\text{g}) \rightarrow 2 \text{Bi}_2\text{O}_3(\text{s})$

$2\text{Bi}_2\text{O}_3(\text{s}) + 12\text{HCl}(\text{aq}) \rightarrow 4\text{BiCl}_3(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$

$4 \text{Bi(s)} + 3 \text{O}_2(\text{g}) + 12 \text{HCl}(\text{aq}) \rightarrow 4 \text{BiCl}_3(\text{aq}) + 6 \text{H}_2\text{O}(\text{l})$

1 Bi : 1 BiCl_3

$(5.00\text{L})(0.200 \text{ mol/L})(1/1)(209 \text{ g/mol}) = 209 \text{ g}$

C12. $\text{pH} + \text{pOH} = 14$, $\text{pOH} = 4.40 = -\log[\text{OH}^-]$ $[\text{OH}^-] = 4.0 \times 10^{-5} \text{ M}$

C13. $+1.24 = +0.80 + E$ $E = +0.44 \text{ V}$ (oxidation), therefore, -0.44 V (reduction)

C15. $\text{Pb}(\text{IO}_3)_2(\text{s}) \rightleftharpoons \text{Pb}^{2+} + 2 \text{IO}_3^-$ $K_{\text{sp}} = 2.6 \times 10^{-13} = [\text{Pb}^{2+}][\text{IO}_3^-]^2 = 4x^3$ $x = 4.0 \times 10^{-5} \text{ M}$

C16. $2 \text{CaSO}_4(\text{s}) \rightleftharpoons 2 \text{CaO}(\text{s}) + 2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g})$

C17. $\ln(0.167) = -(5.20 \times 10^{-3})(t) + \ln(1)$ $t = 344 \text{ s}$

P01-P03 The God Equation, Dr. Michio Kaku, Doubleday, New York, 2021

P01. page 28-30, The second great revolution in technology was the electric age. General Electric backed Thomas Edison and his plan to use DC electricity. Westinghouse backed Nikola Tesla and his plan to use AC electricity. "In the end, Edison lost the battle and the considerable funds he invested in DC technology."

P02. page 46, "Einstein realized that the sun's gravity would be powerful enough to bend the starlight of nearby stars. Since these stars could only be seen during a solar eclipse, Einstein proposed that an expedition be sent to witness the solar eclipse of 1919 to test his theory... When this epic experiment was finally performed by astronomer Arthur Eddington, there was remarkable agreement between Einstein's prediction and the actual result."

P03. page 67-68, in 1930, the Solvay Conference in Brussels featured a debate for the ages. On one side were Albert Einstein and Erwin Schrodinger. On the other side were Niels Bohr and Werner Heisenberg. "Princeton physicist John Wheeler said, 'It was the greatest debate in intellectual history that I know about. In thirty years, I never heard of a debate between two greater men over a longer period of time on a deeper issue with deeper consequences for understanding this strange world of ours.'"

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P04. 21st Century Astronomy, Laura Kay, Stacy Palen, Brad Smith, and George Blumenthal; W.W. Norton; 2013, page 295, Neptune

$$\begin{array}{l}
 \text{P05. } 12 - -9 = 21 \\
 1.00 \times 10^{21}
 \end{array}$$

$$\begin{array}{l}
 \text{P06. } d = \frac{v^2 \sin(2\theta)}{g}, 192 = \frac{v^2 \sin(72^\circ)}{9.8} \\
 v = 44.479560712 \\
 h = \frac{v^2 \sin^2(\theta)}{2g} = \frac{(v)^2 \sin^2(36^\circ)}{2(9.8)} \\
 h = 34.9 \text{ m}
 \end{array}$$

$$\begin{array}{l}
 \sum F_x = ma, T - F_f = ma \\
 T - \mu mg = ma \\
 T - (.333)(4)(9.8) = 4a \\
 \text{P07. } \sum F_x = Ma, F - F_f - T = Ma \\
 F - \mu Mg - T = Ma \\
 30 - (.333)(2)(9.8) - T = 2a \\
 T = 20.0 \text{ N}, a = 1.7366 \text{ m/s}^2
 \end{array}$$

$$\begin{array}{l}
 \omega = \frac{(1800)(2\pi)}{60} = 60\pi \\
 \omega^2 = \omega_0^2 + 2\alpha\theta, \theta = 150(2\pi) = 300\pi \\
 \text{P09. } (60\pi)^2 = 0 + 2\alpha(300\pi), \alpha = 18.849555922 \\
 \omega = \omega_0 + \alpha t, t = \frac{\omega - \omega_0}{\alpha} = \frac{60\pi - 0}{(\alpha)} = 10.0 \text{ s}
 \end{array}$$

$$\begin{array}{l}
 v = f\lambda = (248)(2 \times .65) = 322.4 \\
 \text{P10. } v = \sqrt{\frac{T}{\left(\frac{m}{L}\right)}}, (v) = \sqrt{\frac{T}{0.001036}} \\
 T = 108 \text{ N}
 \end{array}$$

$$\begin{array}{l}
 \rho = \frac{m}{V}, V = \frac{.0004391}{1000} = .0000004391 \\
 \text{P11. } \rho = \frac{.008483}{(V)} = 19,300 \text{ kg/m}^3
 \end{array}$$

$$\begin{array}{l}
 F = \frac{kQ_1Q_2}{r^2} \\
 F_1 = \frac{(8.99 \times 10^9)(100 \times 10^{-6})^2}{(.05)^2} \\
 \text{P12. } \frac{1}{R} = \frac{1}{24} + \frac{1}{48}, R = 16 \\
 R = 16 + 60 + 36 = 112 \\
 \xi = IR, I = \frac{12}{112} = 0.107 \text{ A} \\
 I = 107 \text{ mA}
 \end{array}$$

$$\begin{array}{l}
 F_1 = 35,960 \text{ N } \angle 90^\circ \\
 F_2 = \frac{(8.99 \times 10^9)(100 \times 10^{-6})^2}{(.13)^2} \\
 F_2 = 5,319.5266 \text{ N } \angle 157.38^\circ \\
 F_1 + F_2 = 38,300 \text{ N } \angle 97.4^\circ
 \end{array}$$

$$\begin{array}{l}
 \text{P14. } B = \frac{\mu_0 I}{2\pi r} = \frac{(2 \times 10^{-7})(120)}{2} \\
 B = 0.0000120 \text{ T} \rightarrow 12.0 \mu\text{T}
 \end{array}$$

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$$\xi = -NA \frac{dB}{dt} =$$

$$\text{P15. } \xi = -(100)(120 \times 10^{-4}) \left(\frac{0-.755}{.188} \right)$$

$$\xi = 4.82 \text{ V}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\text{P16. } \frac{1}{24} = \frac{1}{16} + \frac{1}{d_i}, \quad d_i = -48$$

$$M = -\frac{d_i}{d_o} = -\frac{-48}{16} = 3$$

$$.5mv^2 = \frac{hc}{\lambda} - \phi$$

$$\text{P17. } .5(9.11 \times 10^{-31})v^2 = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{240 \times 10^{-9}} - 4.31(1.602 \times 10^{-19})$$

$$v = 550,000 \text{ m/s}$$

$$(.2, .032) \quad (1.2, .188)$$

$$\text{slope} = \frac{.188 - .032}{1.2 - .2} = .156$$

P18. v , neutrino

$$\text{P19. } T = 2\pi\sqrt{\frac{m}{k}}, \quad T^2 = \frac{4\pi^2}{k} \cdot m$$

$$\text{slope} = .156 = \frac{4\pi^2}{k}$$

$$k = 253 \text{ N/m}$$

$$d \sin \theta = m\lambda$$

$$\text{P20. } \sin \theta = \frac{(3)(633 \times 10^{-9})}{\frac{1}{200000}}$$

$$\theta = 22.32129475^\circ$$

$$d = 200 \tan \theta = 82.1 \text{ cm}$$